

Detim Zhao

✉ detimzhao@gmail.com

🌐 [linkedin.com/detimzhao](https://www.linkedin.com/detimzhao)

🐙 github.com/DetimZhao

Education

Bachelor of Science, Software Engineering

May 2026

Arizona State University — GPA: 3.88 / 4.00

Mesa, AZ

- Barrett (Honors), Dean's List (Fall 2022, Spring 2023, Fall 2023, Spring 2024, Spring 2025)

Technical Skills

Languages — Java, Python, C, C++, SQL (PostgreSQL/MySQL), Bash/Shell Scripting

Technologies — pandas, scikit-learn, matplotlib, seaborn, NumPy, PyTorch, Transformers (Hugging Face), OpenCV

Tools — Docker, AWS, Git/GitHub, Unreal Engine 5 (UE5), LaTeX, Jupyter Notebook, UML/PlantUML/Astah

Methodologies — Agile/Scrum, Git Workflows

Experience

Extended Reality (XR) Creative Developer

July 2024 – Present

Meteor Studio

Mesa, AZ

- Containerized GPU-enabled multi-dependency environment using Docker Compose, reducing setup time by 73% (60 to 16 minutes) and eliminating Linux/WSL2 compatibility issues.
- Built volumetric rendering pipeline using RANSAC and ICP algorithms for precise 3D camera calibration in research prototype applications. (Python)
- Developed UE5 prototype of autonomous stereoscopic-camera drone with runner tracking and custom motion-control UI.
- Co-authored two research posters on volumetric rendering and camera calibration, presented at ACM HotMobile 2025.

Undergraduate Researcher

June 2023 – August 2023

Texas A&M-Corpus Christi

Corpus Christi, TX

- Developed multi-drone control system with fellow researcher for PX4 drones in Gazebo simulation, coordinating up to 8-10 autonomous drones via terminal commands. (Python)
- Preprocessed two datasets, then trained and tested five ML models (KNN, SVM, Random Forest, Naive Bayes, Ensemble) for gesture recognition, enhancing classification accuracy by 10%. (Python)
- Co-authored a research paper for publication (DOI: [10.1145/3565287.3617621](https://doi.org/10.1145/3565287.3617621)) that documents findings and work, leading to publication in ACM and conference presentations. (LaTeX)

Projects

Gym Calendar Application (Memoranda) Modernization

- Designed and implemented a proof-of-concept integration with the Google Calendar API, enabling push of gym class schedules and resolving timezone alignment issues.
- Modernized the itinerary UI to improve usability, tailoring agenda views for students and trainers to display only relevant classes and sessions.
- Refactored legacy Java codebase (25,000+ LOC) to eliminate compiler warnings and improve maintainability while managing Git workflows and merge conflicts.

Clustering Analysis of Steam Games Using Reviews, Sentiments, and Metadata

- Conducted clustering analysis on 42,497 Steam games using K-Means and Agglomerative Clustering with custom distance metrics, documenting process and findings in academic paper format. (Python)
- Scraped and analyzed the sentiment of 385 games' English reviews using VADER and vectorized text with BERT embeddings, transforming Steamworks API JSON data into structured formats for clustering and analysis.
- Engineered a modular and reproducible Python workflow for end-to-end data collection, preprocessing, and analysis using pandas, scikit-learn, and transformers, with version control via Git and Git LFS for scalability and efficiency.

Console-Based Chat Application

- Designed PostgreSQL database schema for multi-user chat application with user authentication, message persistence, and real-time data retrieval optimization. (Java, PostgreSQL)
- Developed console-based directions and options for users to register, log in/out, update account info, create a chat room, and etc., streamlining user interactions.